

Shrinkage Baselines for GRPO on GSM8K

Does James–Stein shrinkage of the group mean improve GRPO?

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Abstract

GRPO estimates each prompt’s baseline from the N completions sampled for that prompt alone; with N small this mean is noisy. The assignment proposes replacing it with a James–Stein estimator that shrinks each group mean toward the batch mean. I implemented the specification, six baselines behind one registry, and measured them on Qwen2.5-0.5B-Instruct / GSM8K on a single RTX 5080.

Three findings. **(i)** The specification’s formula applied to raw 0/1 rewards does not work: its $V = 1/N$ assumes the unit per-sample reward variance the specification says to establish first, but 0/1 correctness rewards have $p(1-p) \leq \frac{1}{4}$ and the pseudocode takes the raw matrix. The shrinkage factor collapses and the estimator degenerates into the global mean — and it does so *monotonically in K* , so it fails harder at the batch sizes GRPO is actually deployed at. **(ii)** Estimating V from the data — equivalently, applying the specification after the reward rescaling it assumes — lowers the baseline’s MSE by 16–54% in simulation. The gradient-noise reduction an earlier draft of this report claimed (-21.4% [$-51.5, -7.8$]) came from a projection with a bug and from rollouts drawn with a repetition penalty the policy does not have (Sec. 2); it is withdrawn pending remeasurement. **(iii)** The specification’s optional “divide by the within-group standard deviation” is unsafe in combination with shrinkage, inflating advantages by 10^4 .

I also report a measurement I think matters more than any of the above: two runs of the *same* configuration at the *same* seed end 3.0 accuracy points apart. The effect under test is 0.65–1.5 points. This is why the headline measurement here is gradient noise on paired rollouts rather than end-to-end accuracy.

1 Task

Given `GRPO.md`: replace GRPO’s per-group mean baseline with a James–Stein shrinkage estimator, on GSM8K, with a small policy model. The specification places the change precisely — “after the reward matrix is formed and before advantages are computed” — so what is being rewritten is the *baseline*, not the reward function. [keep this distinction; it was ambiguous in the original brief]

Setup. Qwen2.5-0.5B-Instruct (full fine-tune, no LoRA), GSM8K (7473 train / 1319 test), rule-based verifiable reward (no reward model), one RTX 5080 (16 GB, sm_120). $K = 8$ prompts $\times N = 8$ generations, 512 new tokens, bf16 with gradient checkpointing; 11.0 GiB peak, ~ 25 s/step.

2 Corrections (2026-09-04)

Four defects were found in review after the first draft. All are fixed in the code; none of the numbers below has been re-measured since.

1. **Off-policy rollouts.** `generate()` set temperature, top- p and top- k but not `repetition_penalty`, which fell through to the checkpoint’s `generation_config` — 1.1 for Qwen2.5-Instruct. Completions were sampled from a penalised distribution while the loss used the raw policy’s log-probabilities: off-policy without an importance correction, in every arm of every experiment (training sweep, gradient measurement, evaluation, prompt selection). The loop now passes 1.0 explicitly. Between-arm comparisons remain paired and internally consistent, but they compare baselines under a sampler that is not π_θ .
2. **Projection bug in the gradient measurement.** The Hutchinson probes reused random entries across equal-shaped parameter tensors, so the covariance estimate was invalid. Fixed (continuous stream per probe); Sec. 5 is therefore historical.
3. **Completion mask.** Only `<|im_end|>` counted as a stop token; Qwen2.5 also stops on `<|endoftext|>`, which is its pad token, so such a completion was scored as 512 valid tokens including padding. The mask now stops on every terminator plus the pad id.
4. **Shrinkage diagnostic.** Groups whose mean equals the grand mean were logged as “unshrunk” (1.0) rather than undefined; this is why `global` shows 0.025 in Table 6 instead of its definitional 0.

Beyond the bugs, the noise/signal ratio itself was fragile: $\|\mathbb{E}[g]\|^2$ was obtained as the exact $\mathbb{E}\|g\|^2$ minus the probe estimate of $\text{tr Cov}(g)$, and it is only 8% of $\mathbb{E}\|g\|^2$ here, so a few percent of probe error — which more batches do not reduce — dominates the ratio. The re-measurement will use, alongside the probes, an exact probe-free estimator: for independent batches $\mathbb{E}\langle g_b, g_{b'} \rangle = \|\mathbb{E}[g]\|^2$, so disjoint consecutive pairs of batches give unbiased samples of both $\|\mathbb{E}[g]\|^2$ and $\mathbb{E}\|g\|^2$ (the paper’s own instrument, its eq. 17–18, is the same idea across micro-batches). The training loop now also logs that per-step estimator on request, so the specification’s stability claim can be compared across arms on the same batches; the small- N runs the specification motivates are planned with it.

3 Method

3.1 What the specification says

With group means $X_k = \frac{1}{N} \sum_i r_i^{(k)}$ and grand mean $\bar{X}$,

$$\tilde{\mu}_k = \bar{X} + \text{shrink} \cdot (X_k - \bar{X}), \quad \text{shrink} = \max\left(0, 1 - \frac{c}{S}\right), \quad (1)$$

where $S = \sum_k (X_k - \bar{X})^2$ and $c = V(K-3)$ with $V = \text{Var}(X_k)$ assumed equal to $1/N$. Advantages are $A_i^{(k)} = r_i^{(k)} - \tilde{\mu}_k$.

3.2 The assumption that fails

$V = 1/N$ presumes per-sample reward variance $\sigma^2 \approx 1$. On GSM8K the reward is binary, so $\sigma_k^2 = p_k(1-p_k) \leq \frac{1}{4}$ — at least $4\times$ smaller, and heteroscedastic across prompts (zero for a group that is entirely right or entirely wrong). Overstating V overstates c , which drives shrink to zero.

The failure is not incidental; it is asymptotic in K . Since $\mathbb{E}[S] \approx (K-1) \text{Var}(X_k)$ and $c = V(K-3)$,

$$\frac{c}{S} \xrightarrow{K \rightarrow \infty} \frac{V_{\text{assumed}}}{\text{Var}(X_k)} = \frac{1/N}{\tau^2 + \sigma^2/N}, \quad (2)$$

a constant. Whenever it exceeds 1, shrink $\rightarrow 0$ and the estimator becomes the global mean. For the reward distribution used here ($V_{\text{assumed}} = 0.125$, $\text{Var}(X_k) = 0.074$) the limit is $1.69 > 1$: the collapse is certain, and small K only escapes it by sampling noise in S .

3.3 Baselines compared

name	baseline $b[k, i]$	role
vanilla	group mean	GRPO, the incumbent
rloo	leave-one-out group mean	unbiased variance-reduction control
global	batch mean	ablation: what over-shrinkage degenerates to
js_fixed	Eq. (1), $V = 1/N$	the specification as written
js_pooled	Eq. (1), V estimated	proposed correction
js_loo	two-level leave-one-out shrinkage	Zeng et al. [2]

Table 1: The **global** arm is the one that makes the comparison interpretable: without it, “James–Stein helps” cannot be told apart from “any baseline other than the group mean helps”.

Implementation follows verl’s `register_adv_est` registry so any estimator here drops into that framework through a thin adapter. 162 tests, including a literal transcription of the specification’s ten steps asserted against the implementation to 10^{-12} , and the identity `js_pooled` = specification applied to rewards divided by the pooled within-group standard deviation — i.e. the specification with its own $\sigma^2 \approx 1$ precondition honoured.

3.4 Provenance, and conformance with the listing

Two documents feed the code, in different roles. Everything the policy is trained with comes from `GRPO.md`; the paper of Zeng et al. [2] contributes one optional comparison arm and one diagnostic, neither of which touches the update.

component	source	role
js_fixed	<code>GRPO.md</code> Sec. 3.2–3.3, verbatim	the training path under test
js_pooled	<code>GRPO.md</code> Sec. 3.2 after the Sec. 2 rescaling	the same ten steps on r/σ_{pooled} , scaled back; identity pinned to 10^{-12}
vanilla, global, rloo	standard	GRPO itself, the collapse limit, the unbiased control
js_loo	Zeng et al. eq. 10, 13–16	optional comparison arm; runs only when named
--track-grad-var	Zeng et al. eq. 17–18	per-step gradient variance; a log column, off by default
EBPO [3]	—	cited, not implemented

Table 2: What comes from where. The comparison arm and the variance diagnostic are the only two things taken from the paper.

The listing, run as written. `tests/test_spec_conformance.py` transcribes the ten steps of Sec. 3.2 and, separately, the Python listing of Sec. 3.3 character for character, and runs both against the implementation. Away from the edge cases the listing and `js_fixed` agree to 10^{-12} on continuous and on 0/1 rewards. On its own edge cases the listing does not run:

input	behaviour of the listing
$K < 4$	<code>IndexError</code> : $\bar{X}$ is a 0-d tensor, so <code>unsqueeze(1)</code> is out of range
$S = 0$ (every group mean equal)	<code>TypeError</code> : Python’s <code>max(S, eps)</code> returns the float <code>eps</code> once $S < \varepsilon$, and <code>torch.clamp</code> rejects a float
$K = 8, N = 8, 0/1$ rewards, $S > 0$	runs, and equals <code>js_fixed</code>

Five latent defects in the specification. Each is handled in the code and asserted by a test; none changes the estimator where the specification is well defined.

1. **Step 7 is unreachable.** The prose says “ $S = 0 \Rightarrow$ shrink factor 1”; the listing crashes before reaching it, on exactly the case it claims to handle — and that case occurs at $K = 8$ with 0/1 rewards. We clamp to 0 instead, which is observationally identical: $S = 0$ forces $\text{diff} = 0$, so $\tilde{\mu}_k = \bar{X}$ for any factor.
2. **Step 5 contradicts the listing.** The prose returns the per-group means for $K < 4$; the listing, once its indexing is repaired, returns rewards centred on the *global* mean — the `global` arm, which is no longer GRPO. We follow the prose.
3. **The Sec. 2 precondition is missing from the listing.** The theory assumes rewards rescaled so that $\sigma^2 \approx 1$, which is where $V = 1/N$ comes from; the listing takes the raw matrix. With 0/1 rewards $\sigma^2 \leq \frac{1}{4}$, so c is at least $4\times$ too large: the shrink factor averages 0.2 at $K = 8$ and meets the global mean by $K = 32$ (Sec. 4). `js_pooled` is the listing with the rescaling restored, using the pooled within-group standard deviation because the per-group one is 0 for 30–40% of groups.
4. **The optional “divide by the within-group std” is unsafe under shrinkage.** A degenerate group has std 0 and, by design, a non-zero shrunk advantage, so $A/(0 + 10^{-4})$ scales it by 10^4 (Sec. 6).
5. **Stein’s guarantee does not transfer as stated.** It needs a common, known V and $K \geq 4$; Bernoulli rewards have $V_k = p_k(1-p_k)/N$, different for every prompt and overstated $4\times$ by $V = 1/N$. And a lower-MSE baseline is not an unbiased gradient: the shrunk $\tilde{\mu}_k$ depends on $r_i^{(k)}$ itself, so only the leave-one-out variants (`rloo`, `js_loo`) keep the policy gradient unbiased. The specification’s “more stable, faster convergence, smaller N ” comes with no measurement; the variance diagnostic and the small- N runs of Sec. 2 exist to supply one.

4 Result 1: the specification collapses onto the global mean

Ground truth is unavailable on GSM8K, so this is measured in simulation: $p_k \sim \text{Beta}(1.2, 2.2)$ (mean 0.35, matching the model’s observed per-prompt spread), rewards $\sim \text{Bernoulli}(p_k)$.

K	<code>js_fixed</code> shrink	<code>js_fixed</code> MSE	<code>global</code> shrink	<code>global</code> MSE	<code>js_pooled</code> MSE
4	0.377	+0.4%	0.049	+100.6%	−9.5%
8	0.079	+86.4%	0.023	+118.7%	−18.7%
16	0.016	+121.1%	0.010	+125.0%	−23.8%
32	0.006	+130.8%	0.006	+130.9%	−26.7%
64	0.003	+132.1%	0.003	+132.1%	−28.4%

Table 3: MSE of the baseline against the true p_k , relative to `vanilla`; $N = 8$, 3000 trials. By $K = 32$ `js_fixed` and `global` agree to three significant figures — they are the same estimator. `js_pooled` moves the other way, improving as K grows.

Two consequences worth stating plainly. First, **the specified method gets worse as the batch holds more prompts**, which is the opposite of what more data should buy, and it is the regime GRPO is deployed in. Second, at $K = 4$ it is roughly neutral — so a small- K experiment would show nothing wrong.

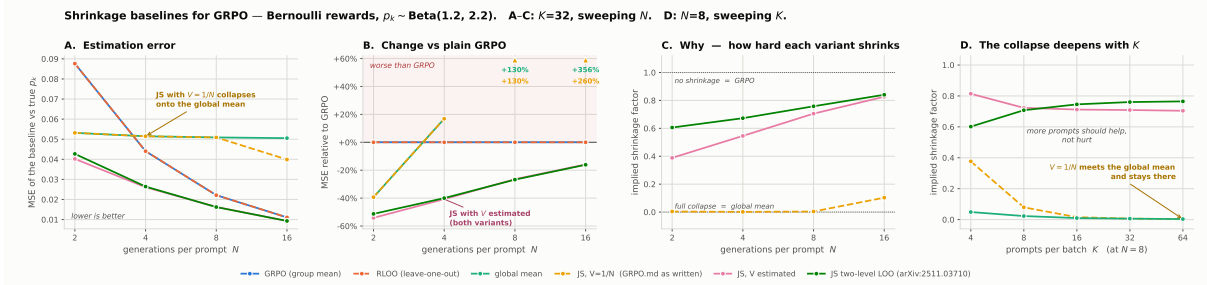


Figure 1: Sweeping N at $K = 32$. `js_fixed` (dashed) lies exactly on global for $N \leq 8$. Estimating V recovers the intended behaviour, and its advantage grows as N shrinks — the regime the method targets.

5 Result 2 (withdrawn): the historical gradient-noise measurement

Every number in this section was produced by the buggy projection and the 1.1 sampler of Sec. 2. It is kept as a record of the instrument, not as a result.

Comparing training runs is the wrong instrument for this (Sec. 7). Instead every baseline scores the *same* rollouts, so rollout randomness cancels and the baseline is the only thing that differs; the policy is frozen, so this measures the estimator rather than a training outcome. The reported quantity is $\text{tr Cov}(g)/\|\mathbb{E}[g]\|^2$ — scale-invariant, because the baselines produce gradients whose magnitudes differ $\sim 5\times$ and raw tr Cov would merely rank them by scale.

baseline	noise/signal	vs GRPO	95% CI	
vanilla	11.5	—	—	
rloo	11.5	−0.3%	[−2.5, +6.9]	no difference (instrument check)
js_loo	9.1	−21.4%	[−51.5, −7.8]	significant
js_pooled	9.9	−14.4%	[−38.2, +0.8]	marginal
js_fixed	6.0	−48.1%	[−81.3, −15.8]	biased — see below
global	5.8	−49.7%	[−83.5, −15.6]	biased — see below

Table 4: Historical, unvalidated: 200 paired batches, $K = 8 \times N = 8$, mean accuracy 29.1%. Intervals are a paired bootstrap over batches, conditional on the 24 probes.

js_fixed and global look best here, and that reading is wrong. Only leave-one-out baselines are independent of the reward they are subtracted from, so only `rloo` and `js_loo` are unbiased for the same gradient. The other two target a *different* $\mathbb{E}[g]$: handing an all-correct group a positive advantage inflates $\|\mathbb{E}[g]\|^2$ with a component that merely pushes up easy prompts. A larger denominator is not more signal.

That `rloo` $\approx$ `vanilla` to within $[-2.5, +6.9]$ is the check that the instrument resolves anything at all — and it agrees with the synthetic study, where leave-one-out changes the baseline’s independence, not its accuracy.

6 Result 3: the standard-deviation variant is a footgun

The specification offers “divide by the within-group standard deviation afterwards” as an option. Under plain GRPO that is safe on a degenerate group only because the advantage is exactly zero there, so $0/(0 + 10^{-4}) = 0$. Every shrinkage baseline gives such a group a *non-zero* advantage by design — that is the point of the method — so the same expression multiplies it by 10^4 .

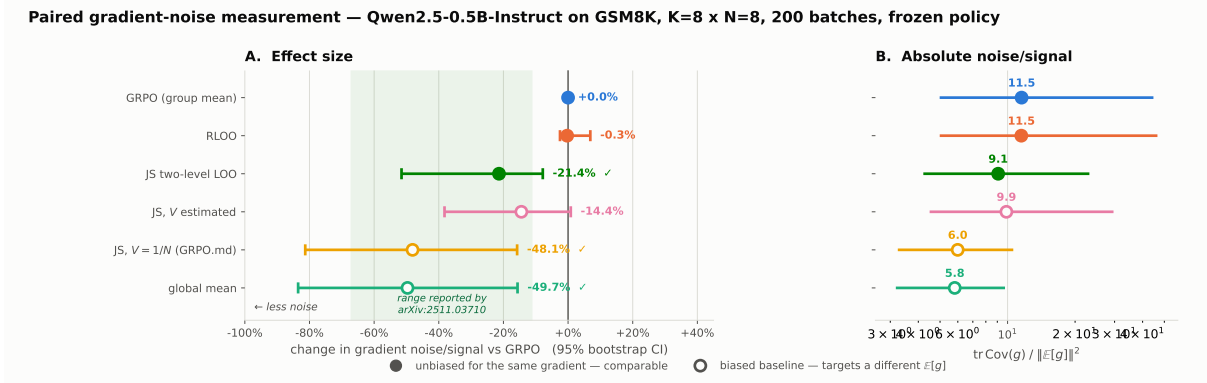


Figure 2: Effect sizes with bootstrap intervals. Shaded: the band reported by Zeng et al. Hollow markers are biased estimators, not comparable at face value.

Measured on GSM8K at $K = 8, N = 8$ with 38% degenerate groups: $\text{Var}(A) = 1.07 \times 10^5$, $\|g\| = 1.7 \times 10^3$. Leaving zero-spread groups unscaled gives 0.57 and 1.2.

7 Why not end-to-end accuracy

	run A	run B
eval@0	42.5%	42.5%
eval@50	44.5%	43.5%
eval@100	44.5%	44.5%
eval@final	48.5%	45.5%

Table 5: Same arm, same seed, same configuration. **eval@0** matches exactly because greedy decoding is deterministic, so the divergence is entirely in training: `torch.manual_seed` does not make sampled generation reproducible on GPU.

The noise floor is 3.0 points. The effect reported for this setting is 0.65–1.5 points (Table 2 of [2]: JS minus GRPO at $N = 8/4/2$, $K = 64$, 500 steps, five seeds) — two to five times smaller, and this report runs $K = 8$ for 150 steps. End-to-end accuracy cannot resolve it on one GPU, and any single-seed accuracy comparison at this scale, in either direction, is an artefact.

8 Training runs

Five arms, 150 steps each, identical seed so every arm walks the same prompt stream (verified: identical reward and degenerate-group fractions at step 0). Every arm evaluates the same 200 test problems in the same order, so checkpoints are compared with McNemar rather than by overlapping binomial intervals.

Nothing here is significant, in either direction. No arm improves on its own starting point at $p < 0.05$, and no arm differs from **vanilla**: against **vanilla**’s final checkpoint, **global** is +2.0 ($p = 0.54$), **js_fixed** +1.5 ($p = 0.68$), **js_loo** +0.0 ($p = 1.00$), **js_pooled** -0.5 ($p = 1.00$). This is exactly what Sec. 7 predicts: the spread across all five arms is 2.5 points, below the 3.0-point floor a single arm shows against itself. **The sweep does not show that shrinkage helps training, and it was not capable of showing it.**

It is worth naming the trap in Table 6: **global** has the largest nominal gain. Read without the p column it says “throw away the group structure entirely and GRPO improves”, which is

arm	eval@0	eval@final	Δ	p	shrink	collapsed
vanilla	42.5%	45.5%	+3.0	0.24	1.000	0/150
js_pooled	42.5%	45.0%	+2.5	0.27	0.850	1/150
js_loo	42.5%	45.5%	+3.0	0.21	0.818	0/150
js_fixed	42.5%	47.0%	+4.5	0.11	0.202	64/150
global	42.5%	47.5%	+5.0	0.06	0.025	133/150

Table 6: p is a paired McNemar of each arm’s final checkpoint against its own **eval@0**. **shrink** is the mean over the last 30 steps; “collapsed” counts steps with shrink < 0.01 . All arms sampled with the 1.1 repetition penalty (Sec. 2); **global**’s 0.025 is the diagnostic artefact described there.

both unsupported and contrary to Sec. 7.

What the sweep does establish is structural. The shrinkage column separates cleanly into three tiers that accuracy cannot resolve but the estimator’s own behaviour makes unambiguous:

vanilla 1.000 $\gg$ js_pooled 0.850, js_loo 0.818 $\gg$ js_fixed 0.202 $\gg$ global 0.025

The two corrected variants retain $\sim 82\text{--}85\%$ of the group structure — mild shrinkage, which is what Stein’s result asks for — while the specified estimator retains 20% and discards it outright on 43% of steps. That **js_pooled** and **js_loo**, derived independently, land within 0.03 of each other is mutual corroboration; it also raises confidence in the $(n - 1)/n$ factor I could not verify against the paper.

At $K = 8$, **js_fixed** is degraded but not identical to **global** (0.202 vs 0.025). Full coincidence needs larger K , as Table 2 shows — which is the deployment regime, not this one.

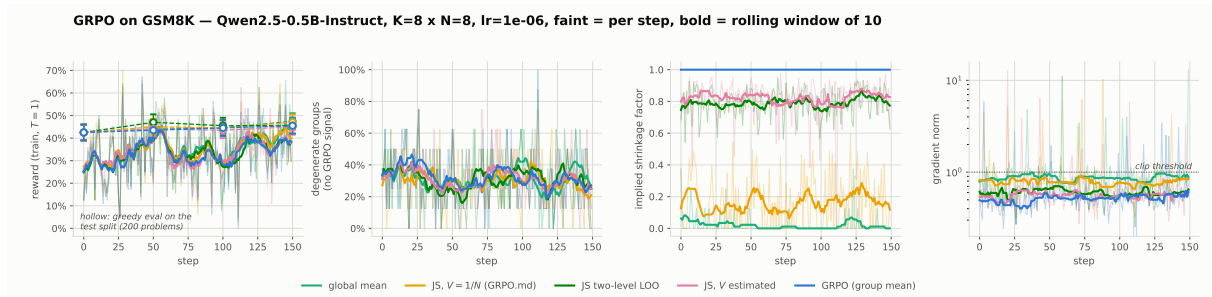


Figure 3: All five arms. Reward and degenerate-group panels overlap almost exactly; the shrinkage panel separates into three tiers. Faint lines are per-step values, bold is a rolling window — median for the gradient norm, which is spiky enough that a rolling mean manufactures flat plateaus that read as stalled training.

9 Limitations

- **No end-to-end accuracy claim.** The sweep confirms this empirically: all five arms sit within 2.5 points, every $p \geq 0.21$. Settling it needs multiple seeds per arm, which one GPU cannot supply.
- **48 batches was not enough** for the gradient measurement — every interval crossed zero. The reported result needed 200. Effect sizes should not be read off short runs here.
- **js_loo is verified** against the arXiv PDF (Sec. 3.3, eq. 10 and 13–16; the $(n - 1)/n$ factor is eq. 15).

- **The specification’s own regime is untested on the GPU.** Its motivation is small N ; every training and gradient run here is $N = 8$, $K = 8$, and no stability metric (clip events, entropy, reward-curve variance) has been compared across arms.
- **Single model, single task.** Whether the collapse threshold moves with reward density is untested.

10 Reproduction

```
uv sync && python main.py smoke          # torch cu129; the RTX 5080 is sm_120
python -m pytest tests/ -q                # 162 tests
python main.py synthetic --k 4 8 16 32 64 --n 8
python main.py gradvar --batches 200
python main.py sweep --baselines vanilla js_fixed js_pooled js_loo global
python main.py compare
```

[repo link] [single seed per arm; a second seed (in progress) would at least bound the run-to-run term] [re-measure everything with the 1.0 sampler before any number above is quoted]

References

- [1] Z. Shao et al. *DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models*. arXiv:2402.03300, 2024.
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